INDIVIDUAL WORK ALLOCATION & DELIVERY PLAN

Multi-Dataset Outbreak Investigation & Evidence Reconstruction Workspace

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Project	OUTBREAK REWIND
Planned workload share	37.5%
Role	Technical Lead, Data Semantics, Evidence Rewind & Integration Owner
Architecture baseline	Revised multi-dataset architecture — 12 November 2026
Purpose	Individual ownership, tasks, deliverables, handoffs and acceptance criteria

Mission

Own technical coherence and the core investigation logic: canonical data semantics, normalized analytical services, provenance, version comparison, Evidence Rewind and end-to-end integration.

Ownership boundary

Owns shared technical contracts, canonical data model decisions, dataset-aware normalization rules, analytics/query semantics, evidence provenance, hypothesis links, version diff, impact preview, revision approval flow and integration. Does not own the frontend or raw file-upload UI.

Workload note: this role carries a larger share of implementation and integration responsibility. Keep interfaces explicit and raise scope changes early rather than silently absorbing another owner's module.

1. Shared non-negotiable project rules

- Keep all original uploaded files unchanged and preserve dataset version, source-row identity and provenance.
- Do not flatten datasets with different grains or metric definitions into one table; retain dataset-specific extensions.
- Missing is not zero. Preserve observed, estimated, unknown and not-applicable distinctions.
- Do not infer person-level cases, contacts or transmission chains from aggregate or outbreak-level sources.
- Every metric and evidence item must be traceable to a dataset version and source record.
- Only compare sources when geography, period, metric definition and units are compatible; surface uncertainty.
- Evidence Rewind proposes deterministic changes and impact previews; a human must approve changes.
- Treat the product as an investigation/research workspace, not a clinical or operational public-health decision system.

2. Detailed task breakdown

1. Technical leadership and shared contracts

- Convert the approved architecture into implementation contracts: identifiers, canonical fields, dataset grain vocabulary, API payloads, status values and error conventions.
- Agree contracts with Anjali Sagar for ingestion/storage and Anjali Kumari for frontend consumption; maintain a decision log and integration checklist.
- Review cross-module changes and keep scope aligned with the two MVP investigation modes.

2. Canonical model, normalization and analytical semantics

- Define canonical shared fields plus dataset-specific extension structures without flattening incompatible sources.
- Own normalization rules for the supplied disease/surveillance and outbreak/event datasets after ingestion, coordinating raw schema mappings with Anjali Sagar.
- Preserve temporal grain and metric semantics across COVID daily, aggregate, age-specific and summary files; keep dengue observed/estimated distinctions.
- Define safe query/aggregation rules for country-day, country-week, country-month, country-year, age-specific and outbreak-level records.
- Implement or coordinate analytics service logic and source-compatible comparisons; ensure missing values are not treated as zero.

3. Evidence provenance and hypotheses

- Implement evidence records linked to investigation, dataset version, source record and relevant normalized/derived records.
- Implement source-traceability services so displayed metrics and evidence can be traced back to source rows.
- Support human-authored hypotheses and links to supporting, contradicting or contextual evidence; persist review status and attribution.

4. Evidence Rewind and integration

- Implement deterministic dataset-version diff outputs: added, changed, removed and unchanged records; flag uncertain matches.
- Track dependencies from source records to normalized records, metrics and evidence links; produce a before/after impact preview.
- Implement approve, reject and request-information decisions; only approved changes create a new investigation revision.
- Integrate frontend, ingestion, analytics, evidence and rewind; add end-to-end tests and a reproducible demo script.

3. Required deliverables

- Architecture implementation notes, canonical data dictionary and API/interface contracts.
- Dataset-aware normalization and analytics/query semantics for both MVP modes.
- Evidence provenance and human-authored hypothesis workflow.
- Deterministic version-diff, impact preview, human decision and revision workflow.
- Integration tests, demo script, technical decision log and clean-checkout setup notes.

4. Definition of done and acceptance criteria

A task is complete only when the implementation is committed, documented and supported by a reproducible test, working endpoint, migration or demonstrated workflow.

- All datasets retain their original grain, source lineage and observed/estimated/missing semantics.
- Metrics are returned only through documented, compatible aggregation rules.
- Evidence details identify exact dataset version and source record.
- Version changes create a deterministic preview; uncertain matches are flagged.
- Only approved changes create a new revision; prior revisions and decisions remain inspectable.
- The end-to-end demo runs from a clean checkout with repeatable tests.

5. Dependencies and handoffs

- With Anjali Sagar: agree raw-file/import outputs, stable dataset/version/source IDs, mapping metadata, validation flags and persistence boundaries.
- With Anjali Kumari: publish API contracts and UI states for investigation, charts, evidence, hypotheses, rewind and history.
- Receive ingestion metadata and quality flags from Anjali Sagar; receive UI payload gaps and interaction tests from Anjali Kumari.
- All three: review interface changes through pull requests and use one shared test investigation.

6. Milestones and sequencing

Sequence is dependency-based. The team should add calendar dates after confirming availability and the review deadline; no dates are invented in this plan.

Milestone	Expected contribution
M0	Lock data semantics, ownership, IDs and demo journeys.
M1	Publish canonical model and API contracts; integrate backend skeleton.
M2	Deliver normalization/analytics plus evidence provenance.
M3	Deliver version diff, impact preview and human approval.
M4	Complete end-to-end integration, tests and demo rehearsal.

7. Explicitly out of scope

- Building the React design system and all frontend pages.
- Owning raw upload, ZIP extraction and initial schema profiling.
- Automated diagnosis, outbreak prediction or AI-generated scientific verdicts.
- Person-level contact tracing or transmission networks unsupported by the datasets.

8. Review/demo contribution

Show dataset-aware analysis with correct metric/time-grain labels, trace one displayed value to its source row, attach evidence to a hypothesis, then upload a changed version and demonstrate the Rewind impact preview and human approval.

9. Weekly progress update

- Completed: PR/commit, endpoint, migration, test result or document link.
- In progress: exact task, status and next action.
- Blocked: dependency, person who can unblock it and decision required.
- Evidence: reproducible command, test output or working demo.
- Risk/change: report scope, security or data-quality risks before implementation.

10. Collaboration and handoff checklist

- Use focused branches and pull requests; request review from affected module owners.
- Do not commit credentials, secrets or private datasets.
- Document setup and sample configuration without secrets.
- Keep field names, identifiers and status values aligned across all workstreams.
- Demonstrate from a clean checkout before marking a feature complete.

11. Shared project completion standard

- One aggregate-surveillance workflow and one outbreak/event workflow work end-to-end.

- Every displayed metric and evidence item can be traced to its dataset version and source record.
- Data limitations and unresolved warnings are visible.
- Rewind changes require human approval and preserve prior revisions.
- Known limitations are documented; mock-ups alone do not count as completed functionality.

Status note: this is a proposed work allocation derived from the revised architecture and supplied dataset inventory. It is not evidence that the listed features have been implemented, scientifically validated, load-tested or security-certified. Verify dataset permissions before deployment.